

DATA MINING AND MACHINE LEARNING (cod. 878II) AY2025-2026 [WAI-LM]

Pandas for Data acquisition, exploration and preprocessing



Version 03-11-2025

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Content of the Lecture

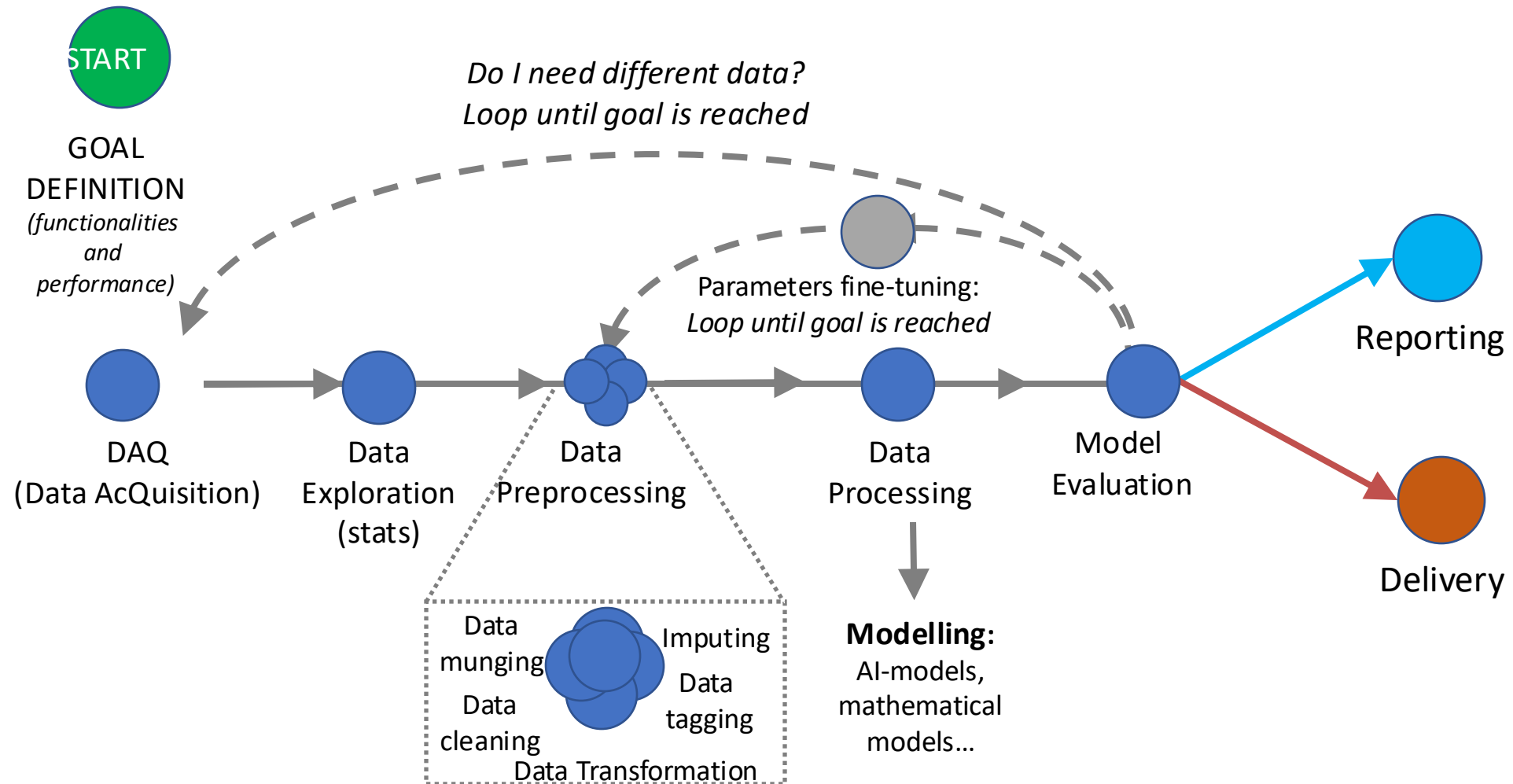
Pandas for:

- ... data acquisition

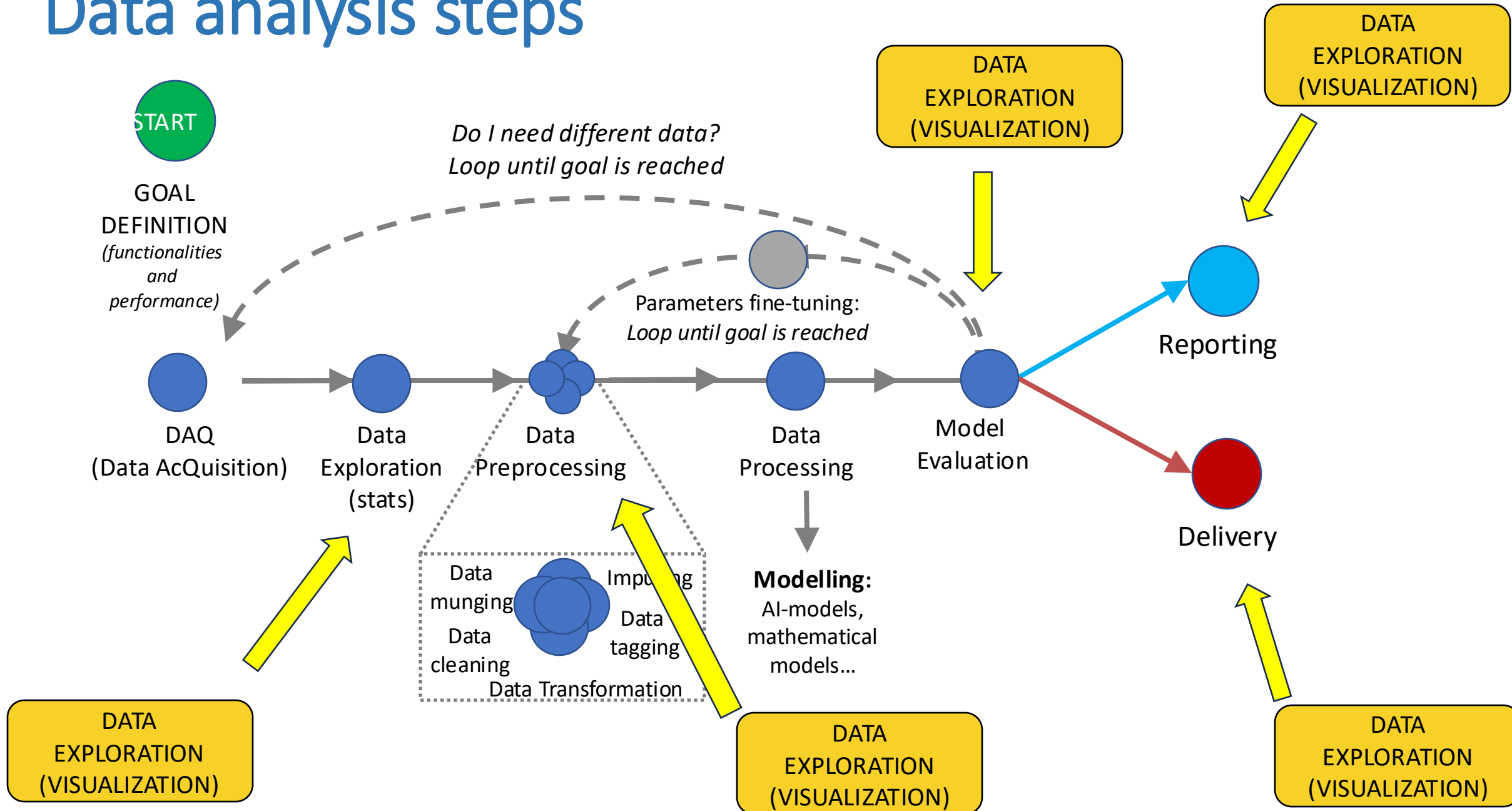
- ... data exploration (numerical and dataviz)

- ... data preprocessing

Data analysis steps



Data analysis steps



Data Acquisition with pandas

Data acquisition with pandas

IN:

Reading data functions

- pd.read_csv
- pd.read_pickle
- pd.read_excel
- pd.read_json
- pd.read_sql
- pd.read_hdf
- ...

```
>> import pandas as pd
>> df = pd.read_csv(filepath_or_buffer="file.csv")
```

df is a pandas DataFrame object (imagine it as a table)

OUT:

Writing functions:

- df.to_csv/_excel/_sql

	Artist	Album	Released	Length	Genre
0	Michael Jackson	Thriller	1982	0:42:19	pop, rock, R&B
1	AC/DC	Back in Black	1980	0:42:11	hard rock
2	Pink Floyd	The Dark Side of the Moon	1973	0:42:49	progressive rock
3	Whitney Houston	The Bodyguard	1992	0:57:44	R&B, soul, pop
4	Meat Loaf	Bat Out of Hell	1977	0:46:33	hard rock, progressive rock

Data Exploration with pandas

Data exploration (numerical) with pandas

- 1.`df.head()`: print out the first 5 rows
- 2.`df.tail(3)`: print out the last 3 rows
- 3.`df.info()` : overview of the dataset -> also: NaN detection
- 4.`df.describe()` #`df.describe(include='all')`: statistical summary
- 5.`df.isna().sum()` <- count NaN values

`df.head()`

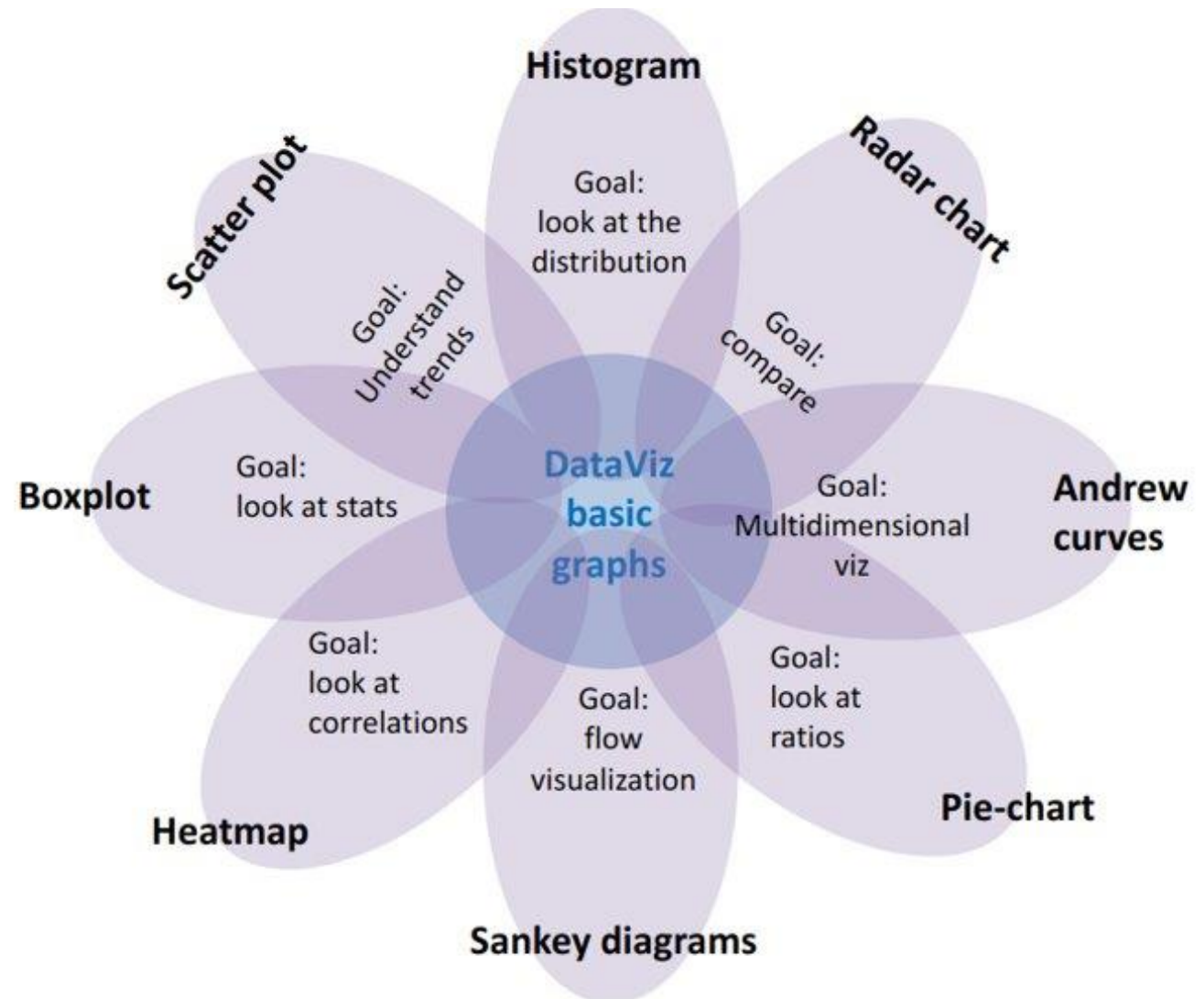
	Artist	Album	Released	Length	Genre	Music Recording Sales (millions)	Claimed Sales (millions)	Released.1	Soundtrack	Rating
0	Michael Jackson	Thriller	1982	0:42:19	pop, rock, R&B	46.0	65	30-Nov-82	NaN	10.0
1	AC/DC	Back in Black	1980	0:42:11	hard rock	26.1	50	25-Jul-80	NaN	9.5
2	Pink Floyd	The Dark Side of the Moon	1973	0:42:49	progressive rock	24.2	45	01-Mar-73	NaN	9.0
3	Whitney Houston	The Bodyguard	1992	0:57:44	R&B, soul, pop	27.4	44	17-Nov-92	Y	8.5
4	Meat Loaf	Bat Out of Hell	1977	0:46:33	hard rock, progressive rock	20.6	43	21-Oct-77	NaN	8.0

Plots for data exploration (not exhaustive)

List of useful plots for data exploration (not exhaustive)

1. Histogram (also: bar chart)
2. Scatter plot (also: line chart)
3. Boxplot
4. Heatmap
5. Pie chart
6. Radar
7. Sankey diagrams
8. Andrew curves

Each of them are useful to look at some characteristics of your dataset. Use the kind of visualization you need for your goal



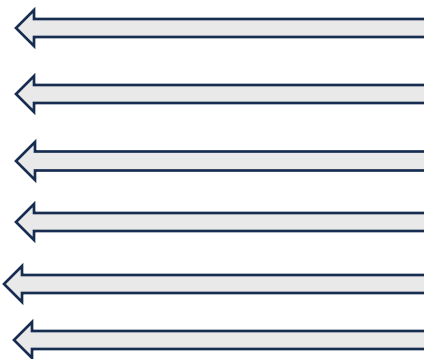
Minimal Dataviz in python

With matplotlib pkg

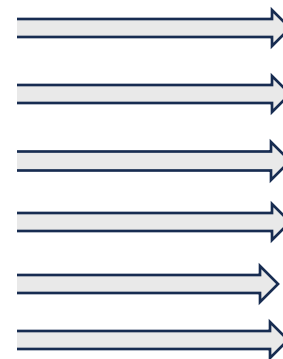
```
import matplotlib.pyplot as plt
import numpy as np
```

```
#given two numpy arrays "x" and "y"
x = np.random.rand(50) # <- shape = (50,)
y = np.random.rand(50)
```

```
plt.plot(x,y)
plt.bar(x,y)
plt.scatter(x,y)
plt.hist(x)
plt.pie(x)
plt.boxplot(x)
```



Line plot
Bar plot
Scatter plot
Histogram
Piechart
boxplot



With Pandas pkg

```
import pandas as pd
import numpy as np
```

```
#given two numpy arrays "x" and "y"
x = np.random.rand(50)
y = np.random.rand(50)
#Given a dataframe "df"
```

```
df = pd.DataFrame({'x':x, 'y':y})
```

```
df.plot(kind='line')
df.plot(kind='bar')
df.plot(kind='scatter', x='x', y='y')
df.plot(kind='hist')
df.plot(kind='pie', x='x', y='y')
df.plot(kind='box', x='x')
```

Plotting a minimal scatter plot with matplotlib

```
import matplotlib.pyplot as plt
import numpy as np

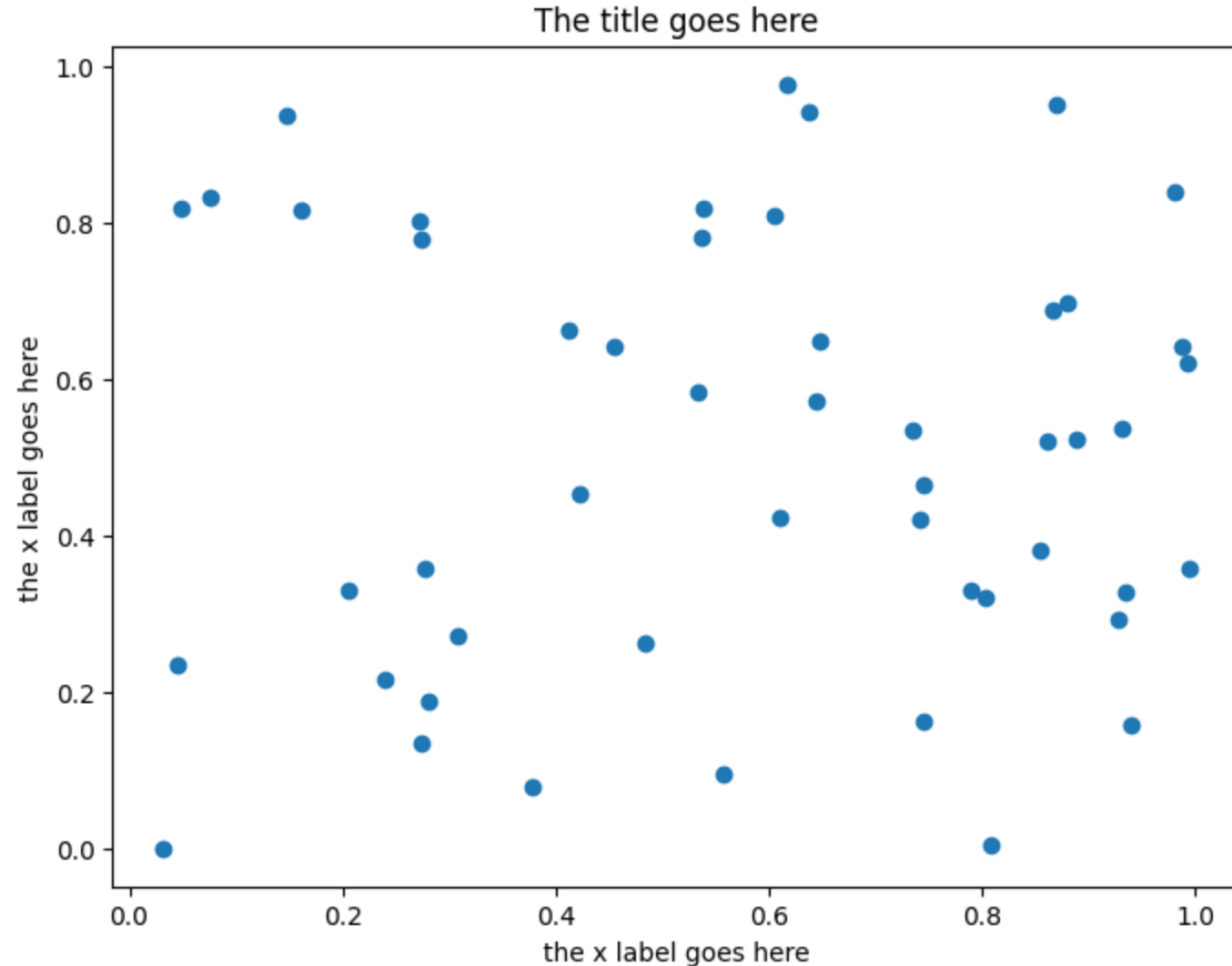
# Generate some sample data
x = np.random.rand(50)
y = np.random.rand(50)

# Create a figure and an axes object
fig, ax = plt.subplots(figsize=(8, 6))

# Plot the data using the axes object
ax.scatter(x, y)

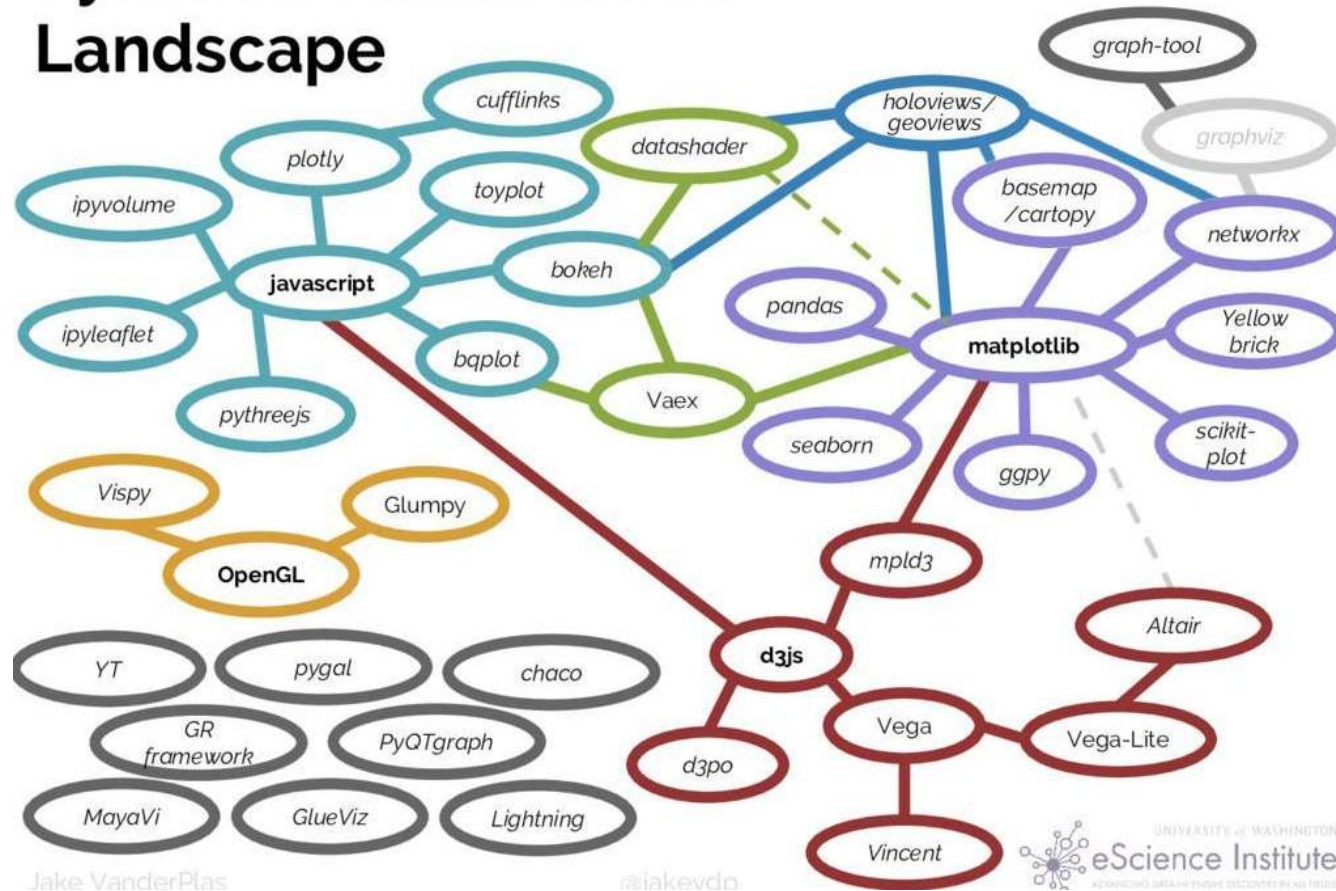
# Set labels and title using the axes object
ax.set_xlabel("the x label goes here")
ax.set_ylabel("the x label goes here")
ax.set_title("The title goes here")

# Show the plot
plt.show()
```



Plots for data exploration (not exhaustive)

Python's Visualization Landscape



- Many different tools, large community of users and developers

Plots for data exploration (not exhaustive) (2)

A "usable" shortlist:

- **matplotlib**: basic package, huge ecosystem (many other packages are built on this)
- **Seaborn**: for statistical-analysis related visualizations, integrates with Pandas
- **Plotly**: for fancy interactive (2D/3D) plots and web easy tools <https://plot.ly/python/> , integrates with Dash
-> also: can be used to have interactive plots during powerpoint presentation
- **Glueviz**: for interaction between plots <http://glueviz.org/>
- **Geodata**:
 - [Shapely](#) -> manipulation and analysis of planar geometric objects (based on widely deployed [GEOS](#))
 - [Geopandas](#) -> Working with geospatial data in Python made easier, combines pandas and shapely.
 - **Folium**: Python + leaflet.js library (maps)
- **Datashader**: for Big data; No interactivity by itself (pairs with Bokeh/Holoviews)
- **Bokeh**: for BIG DATA treating <https://bokeh.pydata.org/en/latest/>
- **Holoviews**: compress bokeh + fancy plots (geoviews + geopandas for geo-referenced data) <https://holoviews.org/>
- **Altair**: interactive, Fast prototyping, reproducible charts

Data exploration: why, how, and when

- **How:**

1. Extract numerical info (focus: missing, data types, statistics such as mean, max, median...)
2. Visualization (a plot is worth 10^3 words)

- **Note: Both numerical and visual are needed!**

- **Why:** to preliminary understand dataset content (the question is: are data adequate for my goal?)

1. How are data distributed? (example: for a classification task, do I have enough data for all classes?)
2. Do I need more/different data? (example: to avoid **gender bias**, do I have enough data to represent both male/female?)

NOTE: WHEN?: Not just at the beginning: after preprocessing, on final results...

(sample) Pearson's correlation coefficient

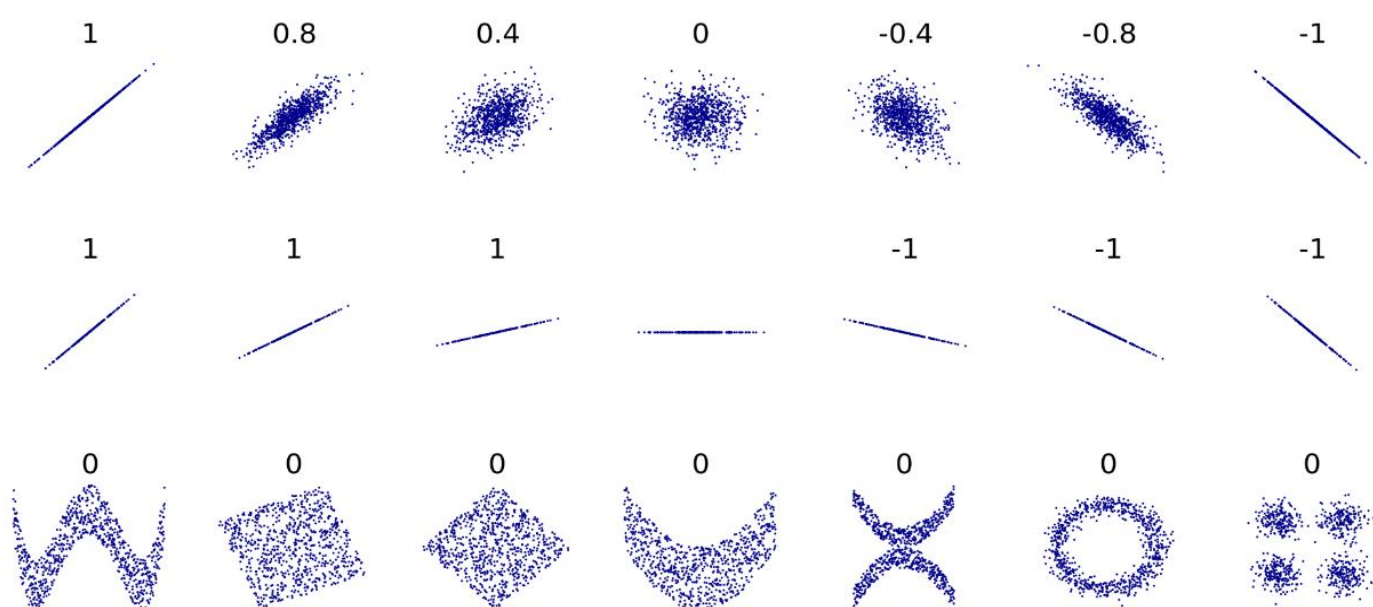
It is common to compute *correlation* to measure the strength of the **linear relationship** between two **numerical** variables.

The (sample) Pearson's correlation coefficient between the variables x and y is given by:

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Where:

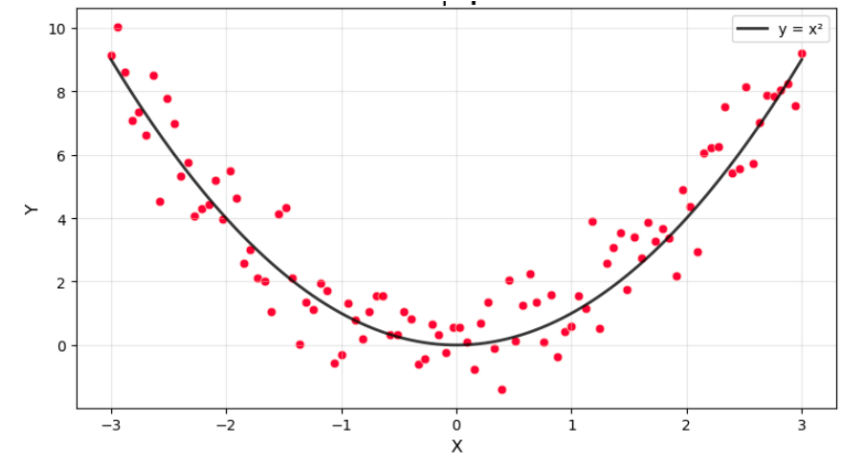
- n is the sample size
- x_i, y_i are the instances at index i
- $\bar{x}$ / $\bar{y}$ is the sample mean



(sample) Pearson's correlation coefficient (2)

PAY ATTENTION:

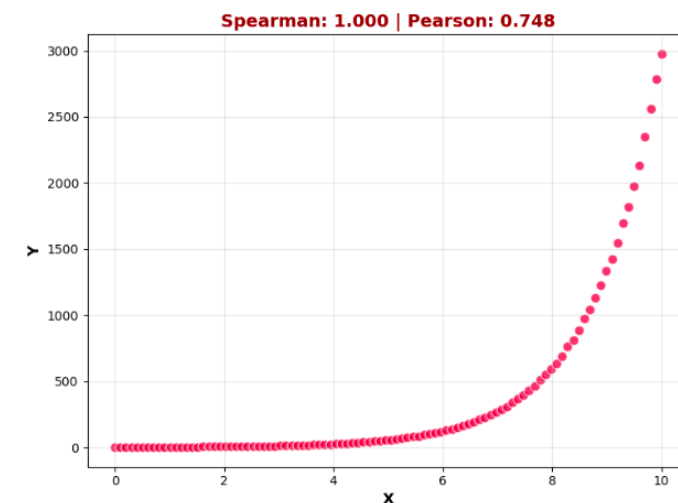
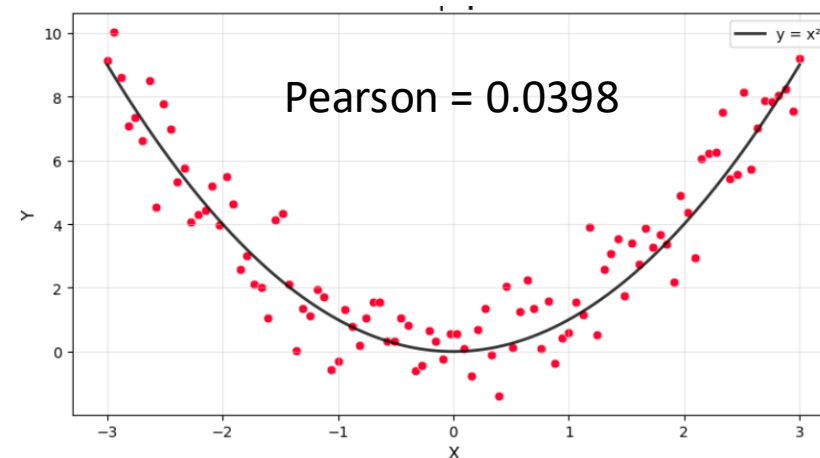
1. Pearson coefficient assume your variables has a **linear relationship**



(sample) Pearson's correlation coefficient (2)

PAY ATTENTION:

1. Pearson coefficient assume your variables has a **linear relationship**



(sample) Pearson's correlation coefficient (2)

PAY ATTENTION:

1. Pearson coefficient assume your variables has a **linear relationship**
2. Pearson correlation **IS NOT for categorical variables**: what is the meaning of a linear relationship between male and female?

Data Preprocessing with pandas

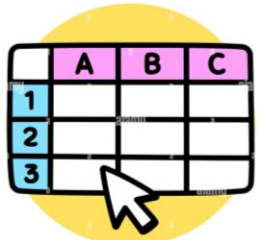
Data preprocessing

1. `df.fillna()`: fill missing
2. `df.dropna()`: remove missing
3. `df.sort_values(by='col_name')`
4. `df.get_dummies()`: convert categorical vars (one-hot encoding)
5. `df = pd.merge(df1, df2)`: merge dataframes
6. `df.groupby(by=["col_name"]).sum()`: group by
- 7....



Note: scikit-learn package used for other preprocessing methods not available in pandas

To summarize



DATA
ACQUISITION

	x	y
count	50.000000	50.000000
mean	0.468157	0.538971
std	0.271167	0.276383
min	0.008343	0.075398

DATA
EXPLORATION
(numerical)



DATA EXPLORATION
(VISUALIZATION)



DATA
PREPROCESSING

Jupyter notebook

Backup

Per la neuroscienza, **quando il cervello si rende conto di un errore inizia a**

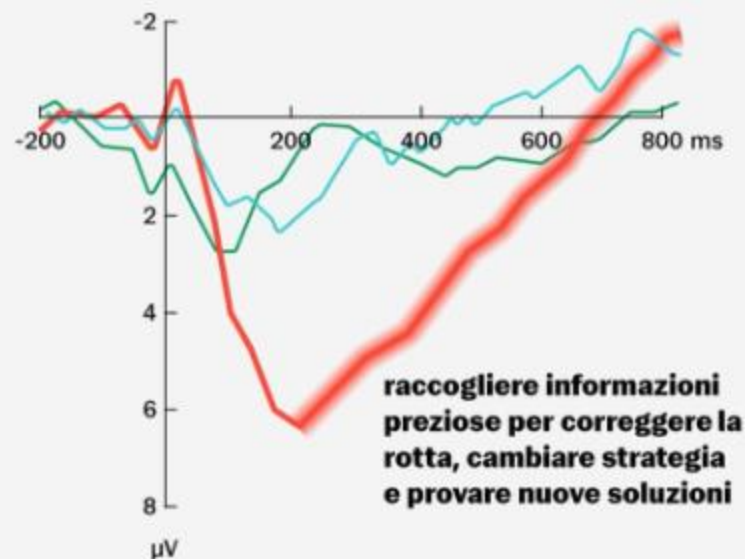


GRAFICO ERP, FRONTO-CENTRAL REGION
FONTE: WANG, L., GUL Y., ZHAO, G. ET AL.
ERROR-RELATED NEGATIVITY AND ERROR AWARENESS
IN A GO/NO-GO TASK. SCI REP 10, 4026 (2020)

torcha.

Cosa emerge dagli studi

Uno studio recente di Microsoft Research e Carnegie Mellon University ("**The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers**", CHI 2025) Microsoft +2

Principali risultati:

- Hanno intervistato 319 "knowledge workers" (professionisti che lavorano con informazioni) e raccolto 936 esempi concreti di uso di GenAI. Microsoft
- Quando le persone hanno **alta fiducia nell'AI**, tendono a esercitare **meno pensiero critico**. Microsoft +1
- Quando le persone hanno alta fiducia nelle proprie capacità, invece, mantengono un livello di pensiero critico maggiore. Microsoft
- Il tipo di pensiero critico che resta attivo cambia: meno dedizione a "pensare da zero", più verifica, integrazione di ciò che l'AI produce, supervisione del risultato. Microsoft +1
- C'è il rischio, segnalato dagli autori, che una dipendenza crescente dall'AI in compiti routinari "tolga" opportunità a esercitare il giudizio, le capacità di problem-solving indipendente, che vengono poi usate solo in casi eccezionali — e questo può portare all'"atrofia" cognitiva (cioè una diminuzione dell'uso regolare di certe capacità). Observer +1